



Universidad
Complutense
Madrid



**Máster Universitario
Ingeniería de Sistemas y de Control**

TRATAMIENTO DE IMÁGENES

Tema 3: TRANSFORMACIONES GEOMÉTRICAS

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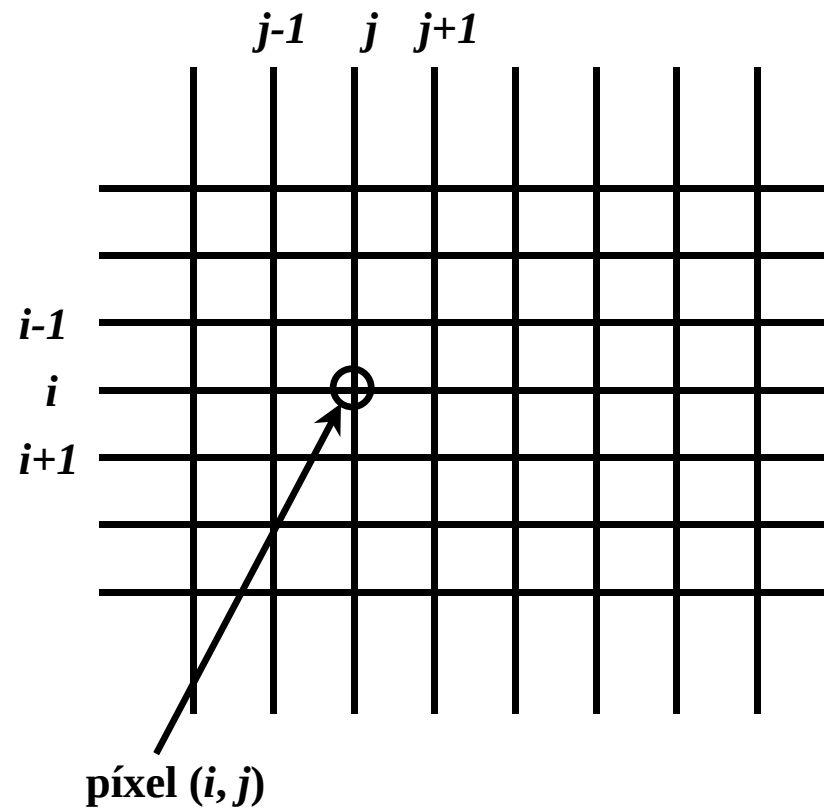
Transformaciones geométricas

- Interpolación
- Transformadas elementales
- Generalización de transformaciones
- Registro



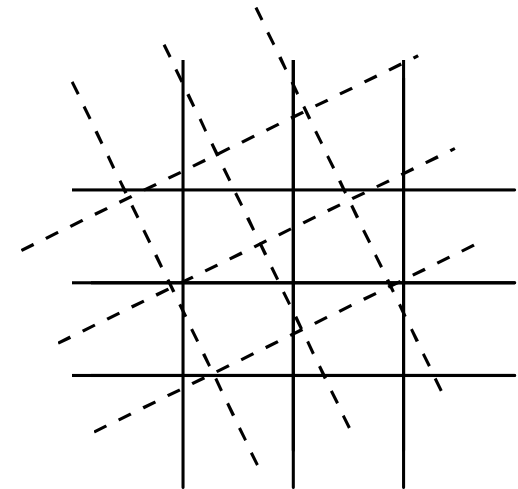
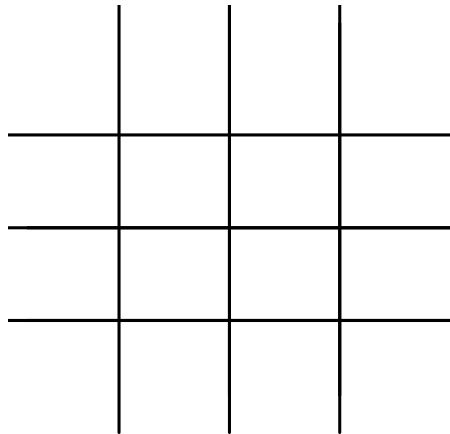


Interpolación

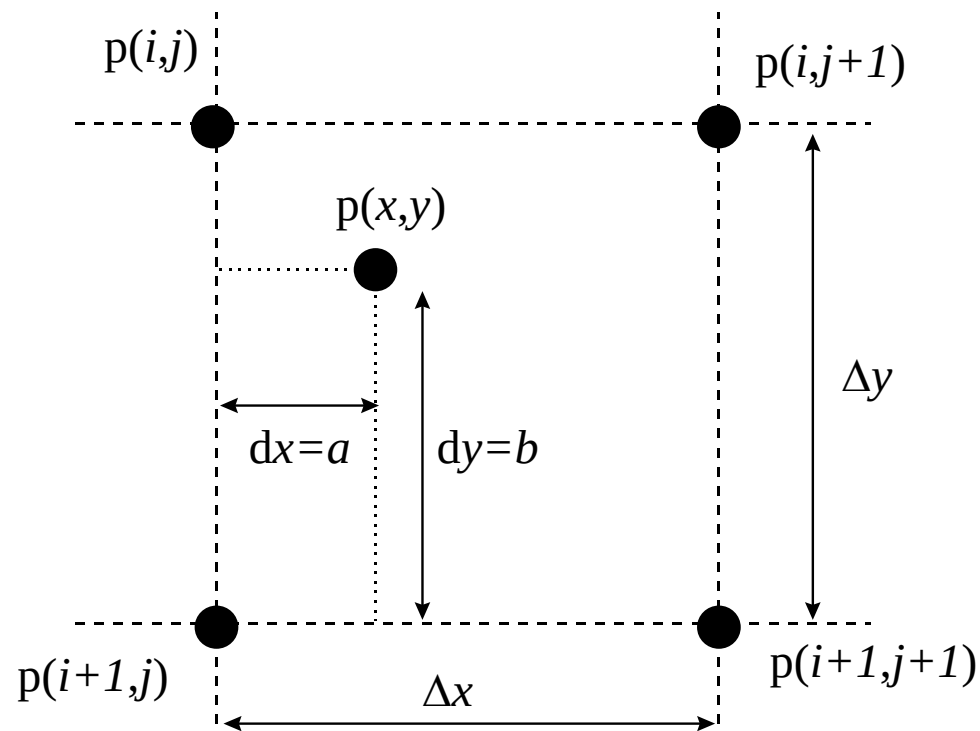




Interpolación



Interpolación



$$p(x, y) = \sum_{i=-n}^n \sum_{j=-m}^m p(i, j) h(x-i, y-j)$$

h : núcleo de interpolación

Interpolación: vecino más próximo

$$h(x) = \begin{cases} 1 & \text{si } 0 < |x| < 0.5 \\ 0 & \text{de otro modo} \end{cases}$$

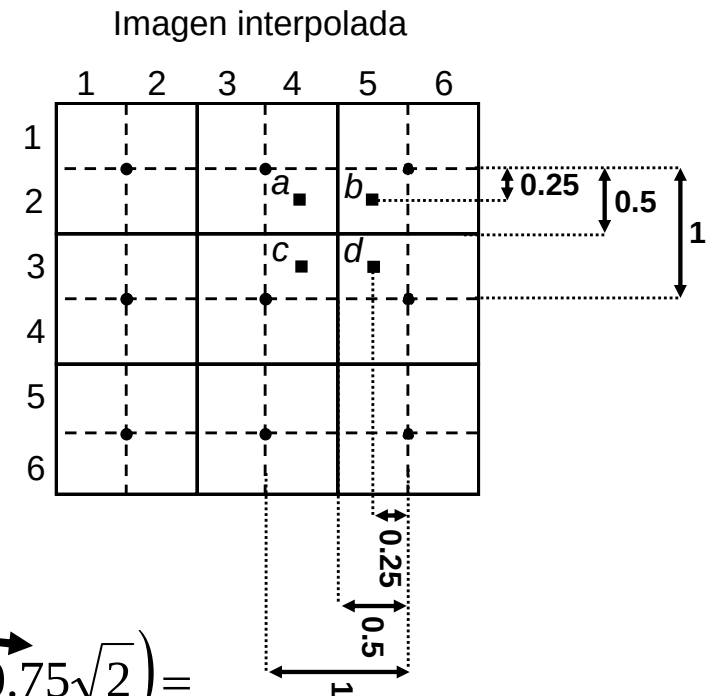
$$p(x, y) = \sum_{i=-n}^n \sum_{j=-m}^m p(i, j) h(x-i, y-j)$$

distancias euclídeas

$$\begin{aligned} p(a) &= 4h(0.25\sqrt{2}) + 8h(0.25\sqrt{10}) + 6h(0.25\sqrt{10}) + 9h(0.75\sqrt{2}) = \\ &= 4h(0.3536) + 8h(0.7906) + 6h(0.7906) + 9h(1.0607) = \\ &= 4 \times 1 + 8 \times 0 + 6 \times 0 + 9 \times 0 = 4 \end{aligned}$$

Imagen original

	1	2	3
1	• 2	• 4	• 8
2	• 3	• 6	• 9
3	• 4	• 1	• 3



Interpolación: **bilineal**

$$a_1 = \left(1 - \frac{dx}{\Delta x}\right) \left(1 - \frac{dy}{\Delta y}\right); \quad a_2 = \frac{dx}{\Delta x} \left(1 - \frac{dy}{\Delta y}\right)$$

$$a_3 = \left(1 - \frac{dx}{\Delta x}\right) \frac{dy}{\Delta y}; \quad a_4 = \frac{dx}{\Delta x} \frac{dy}{\Delta y}$$

$$0 \leq dx \leq 1 \quad 0 \leq dy \leq 1 \quad \Delta x = 1 \quad \Delta y = 1$$

$$a_1 = (1 - dx)(1 - dy); \quad a_2 = dx(1 - dy); \quad a_3 = (1 - dx)dy; \quad a_4 = dxdy$$

$$p(x, y) = a_1 p(i, j) + a_2 p(i, j + 1) + a_3 p(i + 1, j) + a_4 p(i + 1, j + 1)$$

$$h(x) = \begin{cases} 1 - |x| & \text{si } 0 < |x| < 1 \\ 0 & \text{de otro modo} \end{cases}$$

Interpolación: bilineal

$$h(x) = \begin{cases} 1 - |x| & \text{si } 0 < |x| < 1 \\ 0 & \text{de otro modo} \end{cases}$$

$$\begin{aligned} p(a) &= 4h(0.25\sqrt{2}) + 8h(0.25\sqrt{10}) + 6h(0.25\sqrt{10}) + 9h(0.75\sqrt{2}) = \\ &= 4h(0.3536) + 8h(0.7906) + 6h(0.7906) + 9h(1.0607) = \\ &= 4 \times 0.6464 + 8 \times 0.2094 + 6 \times 0.2094 + 9 \times 0 = 4.69 \approx 5 \end{aligned}$$

$$p(b) = 4 \times 0.2094 + 8 \times 0.6464 + 6 \times 0 + 9 \times 0.2094 = 7.89 \approx 8$$

$$p(c) = 4 \times 0.2094 + 8 \times 0 + 6 \times 0.6464 + 9 \times 0.2094 = 6.6 \approx 7$$

$$p(d) = 4 \times 0 + 8 \times 0.2094 + 6 \times 0.2094 + 9 \times 0.6464 = 8.8 \approx 9$$



Interpolación: **bicúbica**

$$h(x) = \begin{cases} 1 - 2|x|^2 + |x|^3 & \text{si } 0 < |x| < 1 \\ 4 - 8|x| + 5|x|^2 - |x|^3 & \text{si } 1 < |x| < 2 \\ 0 & \text{de otro modo} \end{cases}$$

$$\begin{aligned} p(a) &= 4h(0.25\sqrt{2}) + 8h(0.25\sqrt{10}) + 6h(0.25\sqrt{10}) + 9h(0.75\sqrt{2}) = \\ &= 4h(0.3536) + 8h(0.7906) + 6h(0.7906) + 9h(1.0607) = \\ &= 4 \times 0.7941 + 8 \times 0.2441 + 6 \times 0.2441 + 9 \times (-0.0536) = 6.11 \approx 6 \end{aligned}$$

$$p(b) = 4 \times 0.2441 + 8 \times 0.7941 + 6 \times (-0.0536) + 9 \times 0.2441 = 8.9 \approx 9$$

$$p(c) = 4 \times 0.2441 + 8 \times (-0.0536) + 6 \times 0.7941 + 9 \times 0.2441 = 7.5 \approx 8$$

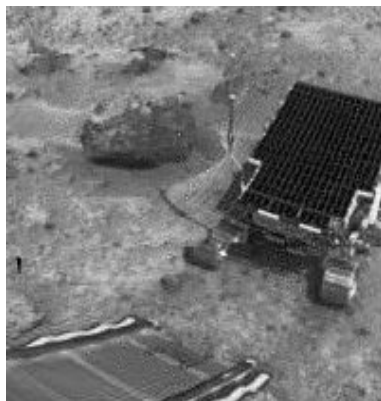
$$p(d) = 4 \times (-0.0536) + 8 \times 0.2441 + 6 \times 0.2441 + 9 \times 0.7941 = 10.3 \approx 10$$

Interpolación

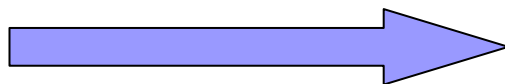
Tema3a.m

interp2

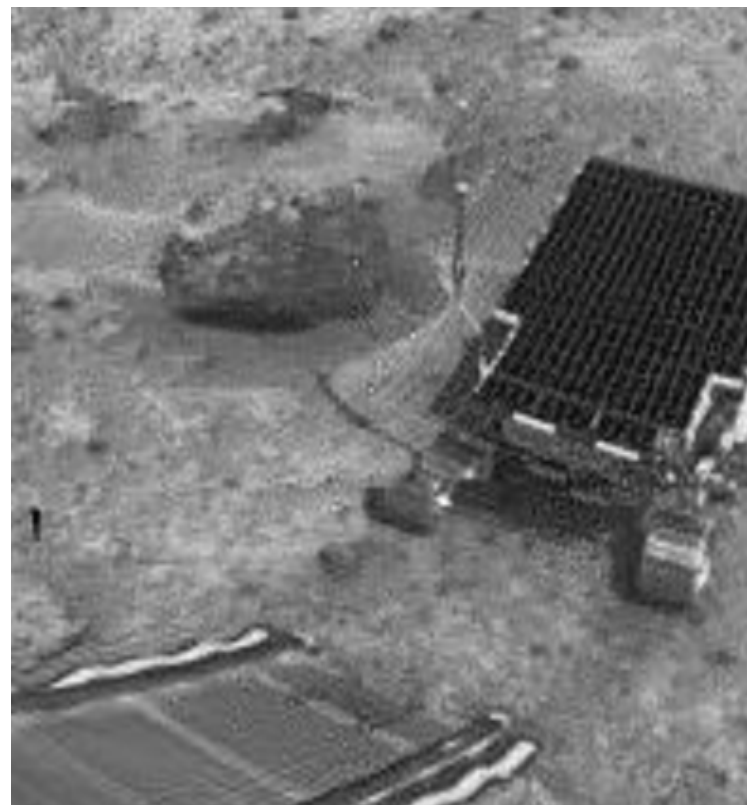
Imagen original



Interpolación bicúbica x 2



Interpolación bicúbica en X e Y x 2





Traslación

$$\begin{aligned}x &= i + i_d \\ y &= j + j_d\end{aligned}$$

$$\begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & i_d \\ 0 & 1 & j_d \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} i \\ j \\ 1 \end{bmatrix}$$

Rotación

$$\begin{aligned}x &= \cos \theta i - \text{sen} \theta j \\ y &= \text{sen} \theta i + \cos \theta j\end{aligned}$$

$$\begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} \cos \theta & -\text{sen} \theta & 0 \\ \text{sen} \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} i \\ j \\ 1 \end{bmatrix}$$

Escalado

$$\begin{aligned}x &= S_x i \\ y &= S_y j\end{aligned}$$

$$\begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} S_x & 0 & 0 \\ 0 & S_y & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} i \\ j \\ 1 \end{bmatrix}$$

Composición de transformaciones

$$\begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & a \\ 0 & 1 & b \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos \alpha & -\operatorname{sen} \alpha & 0 \\ \operatorname{sen} \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -a \\ 0 & 1 & -b \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} c & 0 & 0 \\ 0 & c & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} i \\ j \\ 1 \end{bmatrix}$$



Generalización de transformaciones

$$x = \sum_{p=0}^N \sum_{q=0}^{N-p} a_{pq} i^p j^q$$

$$y = \sum_{p=0}^N \sum_{q=0}^{N-p} b_{pq} i^p j^q$$

Transformada afín

$$\begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} S_x \cos \theta & -\sin \theta & i_d \\ \sin \theta & S_y \cos \theta & j_d \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} i \\ j \\ 1 \end{bmatrix}$$

o equivalentemente

$$\begin{aligned} x &= a_{11}i + a_{12}j + a_{10} \\ y &= a_{21}i + a_{22}j + a_{20} \end{aligned}$$

Otras transformaciones

inclinación:

$$\begin{aligned} x &= i + a_2 j \\ y &= j \end{aligned}$$

perspectiva:

$$\begin{aligned} x &= a_3 ij \\ y &= j \end{aligned}$$

Transformada afín

$$\begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} S_x \cos \theta & -\sin \theta & i_d \\ \sin \theta & S_y \cos \theta & j_d \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} i \\ j \\ 1 \end{bmatrix}$$

$$x = a_{11}i + a_{12}j + a_{10}$$

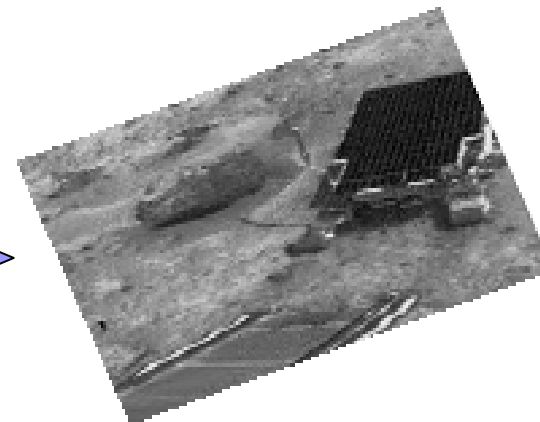
$$y = a_{21}i + a_{22}j + a_{20}$$

$$\theta = -20^\circ, \quad S_x = 1.2 \quad S_y = 0.7 \quad i_d = 50 \quad j_d = -30$$

Imagen original



Imagen transformada afín

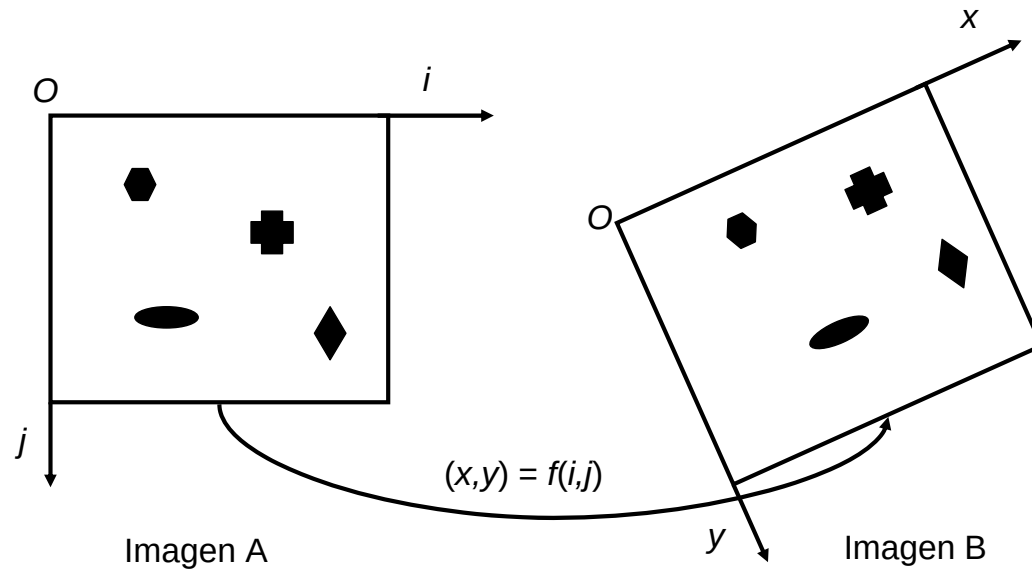


Tema3b.m

En matlab: `imtransform`



Registro entre imágenes



$$C = \begin{bmatrix} 1 & i_1 & j_1 \\ 1 & i_2 & j_2 \\ \vdots & \vdots & \vdots \\ \dots & \dots & \dots \\ 1 & i_n & j_n \end{bmatrix} \quad b_x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \quad b_y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$

Registro entre imágenes

$$C = \begin{bmatrix} 1 & i_1 & j_1 \\ 1 & i_2 & j_2 \\ & \vdots & \\ & \dots & \\ 1 & i_n & j_n \end{bmatrix} \quad b_x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \quad b_y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$

Objetivo: encontrar los parámetros de la transformada (afín)

$$a_x = [a_{10} \quad a_{11} \quad a_{12}]^t \quad a_y = [a_{20} \quad a_{21} \quad a_{22}]^t$$

$$Ca_x = b_x \quad Ca_y = b_y$$

Mínimos cuadrados:

$$(U, S, V) = \text{svd}(C)$$

descomposición en valores singulares

$$a_x = VS^{-1}U^t b_x$$

$$a_y = VS^{-1}U^t b_y$$

verificación
→

$$RMS = \left(\frac{1}{n} \sum_{k=1}^n [(i_k - x_k)^2 + (j_k - y_k)^2] \right)^{\frac{1}{2}}$$

Registro entre imágenes

Encontrar transformada afín (**Ejercicio_03_08.m y Ejercicio_03_09.m**):

$$A = \begin{matrix} & \begin{matrix} -1 & 0 & 1 \\ -1 & 1 & 2 & 3 \\ 0 & 4 & 5 & 6 \\ 1 & 7 & 8 & 9 \end{matrix} \\ \begin{matrix} (i,j) \end{matrix} \end{matrix} \xleftrightarrow{\text{registrar}} A_t = \begin{matrix} & \begin{matrix} 0 & 1 & 2 & 3 & 4 \\ -1 & x & 1 & x & 2 & x \\ 0 & 4 & x & x & x & 3 \\ 1 & 7 & x & 5 & x & 6 \\ 2 & x & x & 8 & 9 & x \end{matrix} \\ \begin{matrix} (x,y) \end{matrix} \end{matrix}$$

$$C = \begin{bmatrix} 1 & i_1 & j_1 \\ 1 & i_2 & j_2 \\ & \vdots & \\ & \dots & \\ 1 & i_n & j_n \end{bmatrix} \quad b_x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \quad b_y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$

$$C^t = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ -1 & 0 & 1 & -1 & 0 & 1 & -1 & 0 & 1 \\ -1 & -1 & -1 & 0 & 0 & 0 & 1 & 1 & 1 \end{bmatrix}$$

$$b_x^t = [1 \quad 3 \quad 4 \quad 0 \quad 2 \quad 4 \quad 0 \quad 2 \quad 3]$$

$$b_y^t = [-1 \quad -1 \quad 0 \quad 0 \quad 1 \quad 1 \quad 1 \quad 2 \quad 2]$$

Registro entre imágenes

Encontrar transformada afín (**Ejercicio_03_09.m**):

$$U = \begin{bmatrix} -0.33 & -0.41 & 0.41 & -0.53 & -0.53 & 0.01 & -0.39 & -0.07 & 0.24 \\ -0.33 & -0.41 & 0 & 0.17 & 0.17 & -0.53 & 0.14 & -0.21 & -0.56 \\ -0.33 & -0.41 & -0.41 & 0.02 & 0.02 & 0.01 & 0.44 & 0.43 & 0.42 \end{bmatrix}$$

$$S^t = \begin{bmatrix} 3 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 2.45 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 2.45 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad V = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix}$$

$$a_x(a_{10}, a_{12}, a_{12}) = VS^{-1}U^t b_x = (2.11, -0.50, 1.67)$$

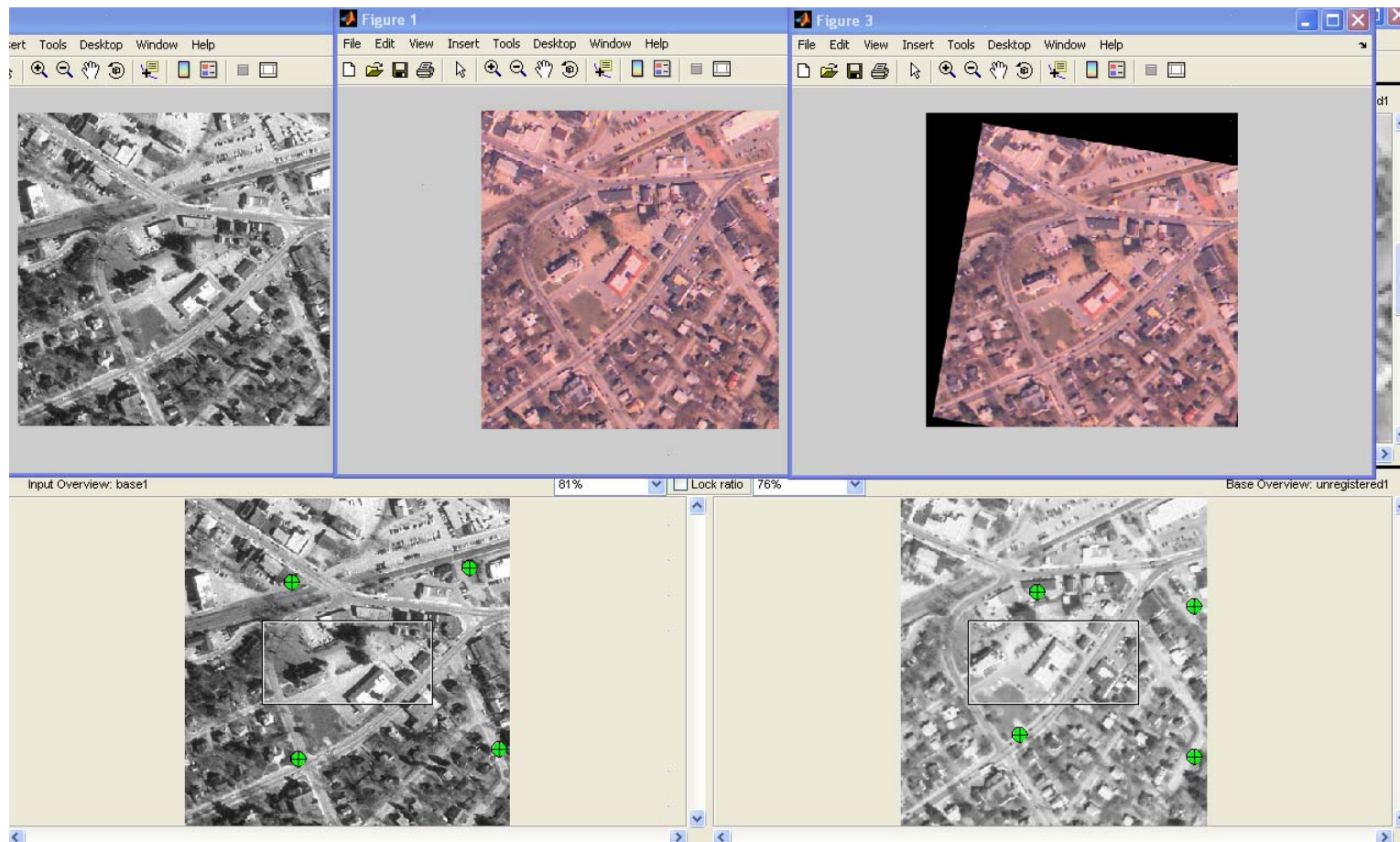
$$a_y = (a_{20}, a_{21}, a_{22}) = VS^{-1}U^t b_y = (0.56, 1.17, 0.50)$$

Registro entre imágenes

Tema3c.m

En matlab:

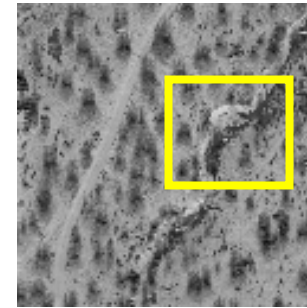
- cp2tform: selecciona puntos de control
- imtransform: aplica la transformada



Registro **automático** entre imágenes

Tema3d.m

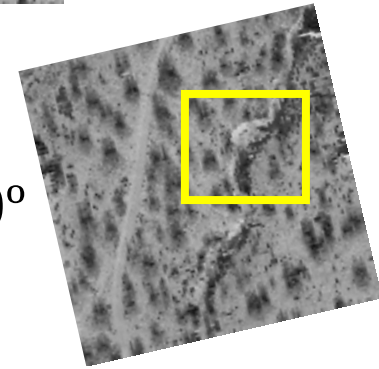
$\theta = 0^\circ$



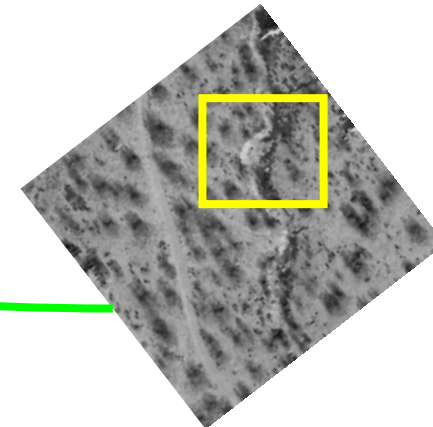
Maximizar Información mutua



$\theta = 20^\circ$



$\theta = 45^\circ$



Puntos de control automáticos



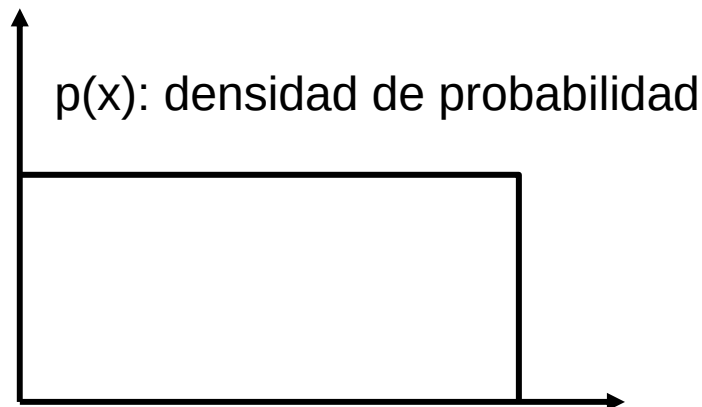
Registro **automático** entre imágenes

Tema3d.m

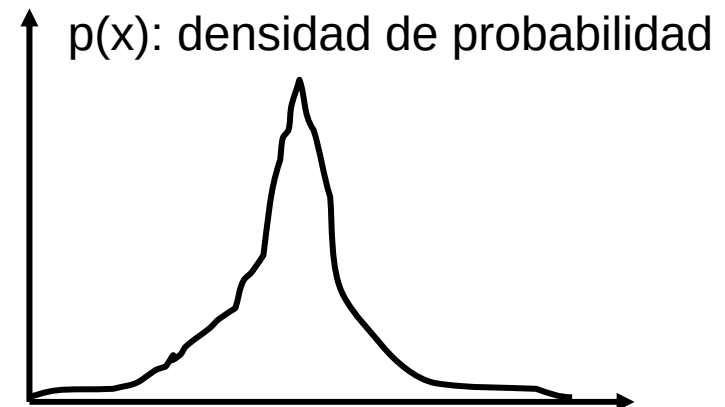
Entropía del histograma:

$$H(X) = -E_X(\log_2 P(X)) = -\sum_{x_i} \log_2(P(X = x_i))P(X = x_i)$$

$$H(X, Y) = -E_X(E_Y(\log_2 P(X, Y)))$$



Alta entropía



Baja entropía

Registro **automático** entre imágenes

Tema3d.m

$$I(X, Y) = H(X) + H(Y) - H(X, Y) \quad \text{estándar}$$

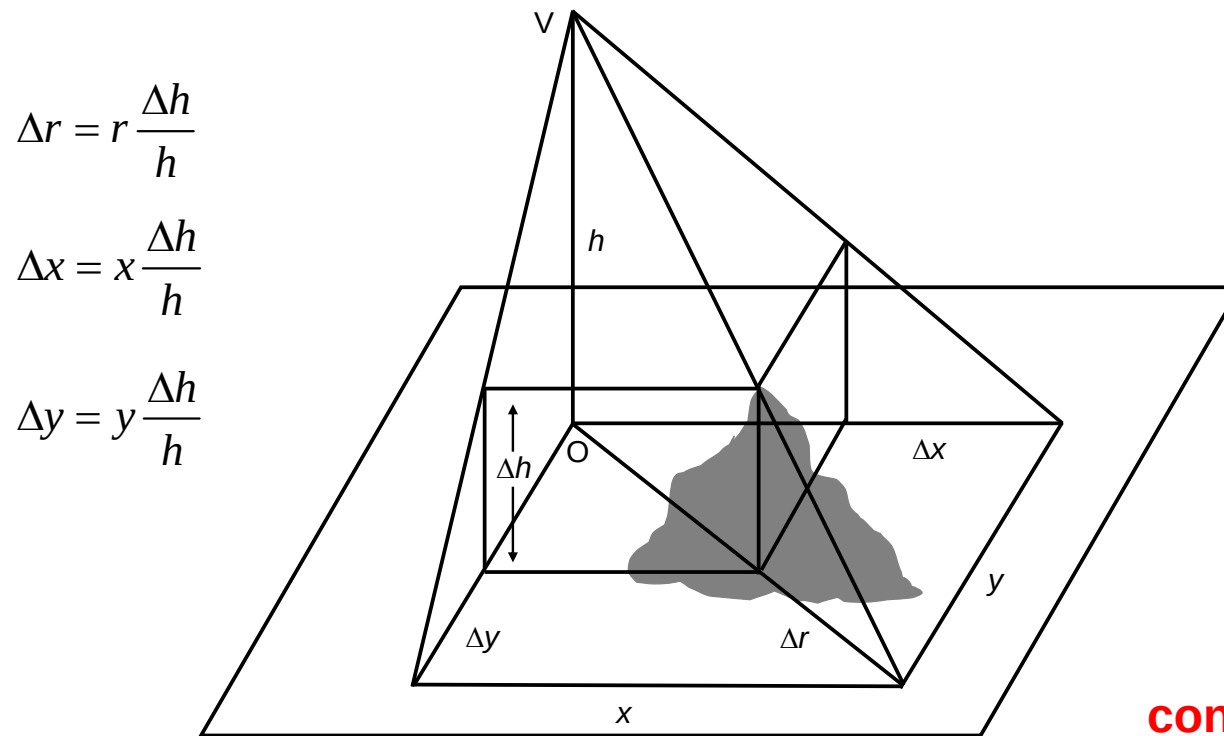
Información mutua:

$$I(X, Y) = \frac{H(X) + H(Y)}{H(X, Y)} \quad \text{normalizada}$$



Corrección por relieve: ortorectificación

Requisitos: Modelo Digital del Terreno (MDT, DTM) ← Estéreo



conocidos

Corrección: → $X = x - \Delta x = x \left(1 - \frac{\Delta h}{h} \right)$ $Y = y - \Delta y = y \left(1 - \frac{\Delta h}{h} \right)$